

Thomas Corbères

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PROFILE

Postdoctoral researcher at the University of Edinburgh, within the Institute of Perception, Action and Behaviour (IPAB). My research focuses on legged locomotion in humanoid and quadruped robots, covering areas such as control, motion planning, and reinforcement learning.

EDUCATION

- 2021-2024* **Ph.D. Informatics, IPAB** The University of Edinburgh
A Complete Framework for Agile Quadruped Locomotion: Integrating Real-Time Control, Planning, and Perception in Multi-Contact Environments, under the supervision of Prof. Steve Tonneau and Dr. Thomas Flayol.
- 2016-2020* **French Graduate School of Aeronautics and Space, ISAE-SUPAERO**
A selective three year program leading to a French master's degree.
- Computer science, Applied Mathematics, Physics.
 - Major : Signals and Systems. Estimation, Advanced control engineering.
 - Field : Autonomous Systems. Robotics, Navigation, Perception.
 - Big Data course (Toulouse Tech) Statistics, Machine Learning.
- 2014-2016* **Preparatory classes, Montaigne (Bordeaux)**
Two-year intensive program in mathematics, physics, and engineering, preparing for national competitive exams to enter French graduate schools "Grandes Écoles".

EXPERIENCE

- 2024-2025* **The University of Edinburgh**, Postdoctoral researcher
Under the supervision of Steve Tonneau. This part of my research focuses on differentiable reinforcement simulators, contact-implicit simulations, and reinforcement learning.
- 2020* **LAAS-CNRS**, 6 months internship (Toulouse, France)
End-of-studies internship. Implementation of a Model Based Predictive Controller for a quadruped robot using a reduced model dynamics.
- 2019* **THALES**, 6 months internship (Mérignac, France)
Internship within the Thales Innovation Laboratory. Contributed to the prototyping of an avionics system page for the FlytX touchscreen cockpit using Qt/QML.
- 2018* **AIRBUS D&S**, 5 months internship (Toulouse, France)
Internship within the the SCAO department -Attitude control system and Orbit-. Conducted a study on noise transmission in the control loop of an E2000+ family satellite.

PUBLICATIONS

- [1] Thomas Corbères, Thomas Flayols, Pierre-Alexandre Léziart, Rohan Budhiraja, Philippe Souères, Guilhem Saurel, and Nicolas Mansard. "Comparison of predictive controllers for locomotion and balance recovery of quadruped robots". In *2021 IEEE International Conference on Robotics and Automation (ICRA)*, pages. 5021-5027, 2021.
- [2] Fanny Risbourg¹, Thomas Corbères¹, Pierre-Alexandre Léziart, Thomas Flayols, Nicolas Mansard, and Steve Tonneau. "Real-time footstep planning and control of the solo quadruped robot in 3d environments". In *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 12950-12956, 2022. ¹Joint first authors
- [3] Pierre-Alexandre Léziart, Thomas Corbères, Thomas Flayols, Steve Tonneau, Nicolas Mansard, and Philippe Souères. Improved control scheme for the solo quadruped and experimental comparison of model predictive controllers". *IEEE Robotics and Automation Letters (RA-L)*, 7(4):9945-9952, 2022.
- [4] Carlos Mastalli, Saroj Prasad Chhatoi, Thomas Corbères, Steve Tonneau, and Sethu Vijayakumar. Inverse-dynamics mpc via nullspace resolution". *IEEE Transactions on Robotics (TRO)*, 39(4):3222-3241, 2023.
- [5] Jaehyun Shim, Carlos Mastalli, Thomas Corbères, Steve Tonneau, Vladimir Ivan, and Sethu Vijayakumar. Topology-based mpc for automatic footstep placement and contact surface selection. In *2023 IEEE International Conference on Robotics and Automation (ICRA)*, pages 12226-12232, 2023.
- [6] Thomas Corbères, Carlos Mastalli, Wolfgang Merkt, Ioannis Havoutis, Maurice Fallon, Nicolas Mansard, Thomas Flayols, Sethu Vijayakumar, and Steve Tonneau. "Perceptive locomotion through whole-body mpc and optimal region selection". *IEEE Access* 2024.
- [7] Gianni Lunardi, Thomas Corbères, Carlos Mastalli, Nicolas Mansard, Thomas Flayols, Steve Tonneau, and Andrea Del Prete. Reference- free model predictive control for quadrupedal locomotion. *IEEE Access*, 12:689-698, 2024.
- [8] Thomas Corbères. A Complete Framework for Agile Quadruped Locomotion: Integrating Real-Time Control, Planning, and Perception in Multi-Contact Environments. PhD thesis, The University of Edinburgh, 2024.

SKILLS

Programming

OS : Linux, Windows
Languages : Python, C++, Jax, PyTorch

Languages

English : Fluent
Spanish : Level B1